

PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR
CAT- 2 (2023-2024)

Department :- Computer Technology
Subject :- Data Analytics using Python
Duration : 1.5 Hrs

Semester :- VIII A/B
Subject Code :- BECT802T
Max Marks :- 35

Note: 1) All Questions are Compulsory
2) All questions carry marks as indicated

Q. No.	Questions	Marks	CO	BL
Q 1 A) I.	Which of the following is true about multiple regression model? a) It has only one independent variable b) It has more than one dependent variable c) It has more than one independent variable d) It has at least 2 dependent variable	1 M	CO3	1
II.	The interval estimate of the mean value of y (dependent variable) for a given value of x is defined as? a) Prediction interval estimate b) Confidence interval estimate c) Average regression d) X vs Y correlation interval	1 M		1
B)	What is multiple regression? Explain the estimation process For multiple regression.	5 M		2
OR				
Q 2 A) I.	What is the range of values as output from the sigmoid function? a) 0 to 1 b) -1 to -1 c) -1 to 0 d) None of these	1 M	CO3	1
II.	In logistic regression, a) The graph doesn't follow S shape curve b) The dependent variable is categorical c) The estimated value of dependent variable is not probability d) None of the above.	1 M		1
B)	What do you mean by logistic regression and explain when it is used.	5 M		2
Q 3 A) I.	Sensitivity in ROC analysis is defined as: (Here, TP : True Positive, TN: True Negative, FP: False Positive, FN: False Negative) a) $FP / (FP + TN)$ b) $FN / (TP + FN)$ c) $TN / (TN + FP)$ d) $TP / (TP + FN)$	1 M	CO4	1
II	State true or False: Precision is inversely proportional to recall a) True b) False	1 M		1
B)	What is ROC? Define Sensitivity in ROC analysis	5 M		1
C)	Describe various Accuracy Measures for Classification.	7 M		2

OR

- Q 4 A) I. Sampling distribution for a goodness of fit test is the
 a)Poisson distribution
 b)t distribution
 c)Normal distribution
 d)Chi-square distribution
 1 M CO4 1
- II. Statistical test conducted to determine whether to reject or not reject a hypothesized probability distribution for a population is known as a _____
 a)Contingency test
 b)Probability test
 c)Goodness of fit test
 d)None of these alternatives is correct
 1 M 1

B) What do you mean by cluster analysis ? State its purpose.
 5 M 1

C) Describe the metrics for computation of distances between the objects.
 7 M 2

- Q 5 A) I. Which of the following can act as possible termination conditions in K-Means?
 a)For a fixed number of iterations.
 b)Assignment of observations to clusters does not change between iterations.
 Except for cases with a bad local minimum.
 c)Centroids do not change between successive iterations.
 d)Terminate when RSS falls below a threshold.
 1 M CO5 1

- II. Which library is used for calculating distance measures in clustering using python?
 a)distance_matrix
 b)scipy.spatial
 c)scipy_spatial
 d)distance.matrix
 1 M 1

B) State the classification of clustering methods.
 5 M 1

C) Explain K-means clustering algorithm.
 7 M 2

OR

- Q 6 A) I. _____ is the measure of uncertainty of a random variable, it characterizes the impurity of an arbitrary collection of examples.
 a)Information Gain
 b)Gini Index
 c)Entropy
 d)None of the above
 1 M CO5 1

- II. Which library is used to build the decision tree model-
 a)Decision tree classifier
 b)DecisionTreeClassifier
 c)Decision_Tree_Classifier
 d)Decision_tree_model
 1 M 1

B) What is regression analysis? State Simple Linear Regression Model.
 5 M 1

C) What are different attribute selection measures in decision tree? Explain with example.
 7 M 2